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Mental imagery in the navigation domain: a computational model of sensory-motor simulation mechanisms

Fabian Chersi¹, Francesco Donnarumma¹ and Giovanni Pezzulo^{1,2}

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Abstract

Recent experimental evidence indicates that animals can use mental simulation to make decisions about the actions to take during goal-directed navigation. The principal brain areas found to be active during this process are the hippocampus, the ventral striatum and the sensory-motor cortex. In this paper, we present a computational model that includes biological aspects of this circuit and explains mechanistically how it may be used to imagine and evaluate future events. Its most salient characteristic is that choices about actions are made by simulating movements and their sensory effects using the same brain areas that are active during overt execution. More precisely, the simulation of an action (e.g., walking) creates a new sensory pattern that is evaluated in the same way as real inputs. The model is validated in a navigation task in which a simulated rat is placed in a complex maze. We show that hippocampal and striatal cells are activated to simulate paths, to retrieve their estimated value and to make decisions. We link these results with a general framework that sees the brain as a predictive device that can ‘detach’ itself from the here-and-now of current perception using mechanisms such as episodic memories, motor and visual imagery.

Keywords

Mental simulations, hippocampus, neural network, sensory-motor transformations, forward sweeps, ventral striatum

1 Introduction

A series of studies by Tolman (1948) revealed that rats have the ability to flexibly adapt their navigation strategies and plan novel routes to rewarding sites when old ones are prevented. To explain these findings, Tolman argued that rats form ‘cognitive maps’ of their environment and use them to plan and foresee action possibilities, rather than relying on stimulus-response learning. More recent studies are starting to reveal the neural underpinnings of these abilities. In particular, evidence is accumulating that rats can mentally simulate future trajectories, an ability linked to so-called ‘forward sweeps’ in the hippocampus (Dragoi & Tonegawa, 2011; Gupta, van der Meer, Touretzky, & Redish, 2010; Johnson & Redish, 2007; Pastalkova, Itskov, Amarasingham, & Buzsáki, 2008; van der Meer & Redish, 2009, 2010) (Figure 1). It has been hypothesized that mental simulations are used for goal-directed decision-making, as they permit to generate reward expectations linked to the different behavioural alternatives (Lisman & Redish, 2009; Pennartz, Ito, Verschure, Battaglia, & Robbins, 2011; van der Meer & Redish, 2009). This view is reminiscent of the idea of ‘vicarious trial and error’ (VTE) originally proposed by Tolman

(1948) as an exploration mechanism and more recently studied as a mechanism of search and goal-directed choice (Papale, Stott, Powell, Regier, & Redish, 2012; Redish, 2012).

Although it is still unclear how flexible mental simulations are (e.g., if they can really apply to novel scenarios or if they are a replay of old scenarios), a certain degree of flexibility has been demonstrated, such as the ability to simulate shortcuts (Gupta et al., 2010). This evidence, together with the strong similarities between the brain systems for remembering the past and imagining the future, supports the idea that episodic memories can be flexibly linked to future events and reassembled in novel ways (Hassabis, Kumaran, & Maguire, 2007; Schacter et al., 2012).

Although the kind of mental imagery we analyse here is linked to spatial navigation, it is an example of a more general class of abilities that humans and other

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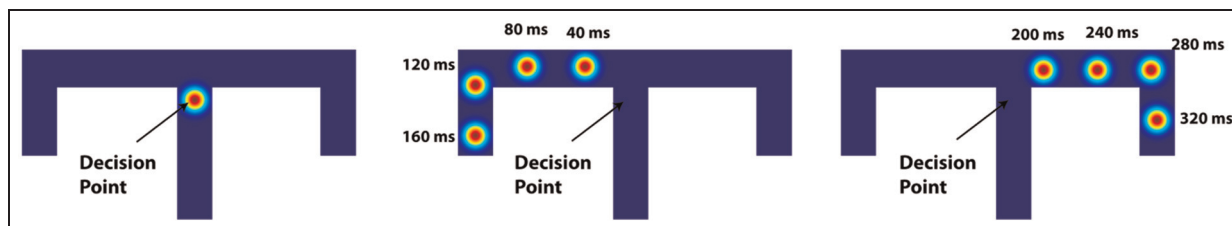


Figure 1. Sequence of place representations decoded from actual neurons as the rat pauses at a decision point of a T-crossing within a maze (location indicated by the arrow). Note that even though the rat remains stationary (but can move the head), the decoded position sweeps down first one arm of the maze (central panel), then the other one (right panel). Adapted from Johnson and Redish (2007).

animals use to self-generate stimuli and flexibly ‘detach’ from the here-and-now of current perception to foresee future events (Pezzulo & Castelfranchi, 2007). These abilities have been studied in separate domains (e.g., mental simulation in relation to episodic memory, visual imagery in perceptual domains, motor imagery in the domains of motor control and skill learning) and associated to different brain areas (e.g., hippocampus, visual or motor cortex) but they could all use similar mechanisms for the re-enactment of stored sensory-motor patterns and prediction of action outcomes. Furthermore, taken together they characterize the brain as a *predictive device* that continuously strives to foresee future events and generate future action possibilities to better adapt to an uncertain and dynamic world (Bar, 2009; Moulton & Kosslyn, 2009; Pezzulo, Butz, Castelfranchi, & Falcone, 2008; Schacter et al., 2012). Understanding the nature and functioning of mental simulations in one domain can shed light on the more general class of mechanisms devoted to ‘future thinking’ in the human and animal brains.

2 Model hypothesis

The model we propose in this paper aims at providing a biologically realistic explanation of how mental simulation applied to spatial navigation problems may work in animals. The model focuses on the neuronal circuit formed by the hippocampus, the ventral striatum (VS) and the sensory-motor cortex, which (as discussed above) plays a key role in goal-directed navigation. The fundamental assumptions of the proposed model can be summarized as follows.

1) Hippocampal place cells are modulated by spatial and sensory information. This has been demonstrated by a large amount of experiments that show that place cells in the hippocampus respond to mostly unique spatial locations (Muller, Kubie, Bostock, Taube, & Quirk, 1991; O’Keefe & Burgess, 2005; O’Keefe & Dostrovsky, 1971). In the current implementation, for the sake of simplicity, hippocampal neurons respond only to sensory input and

their activation indicates that the agent is located inside the receptive field of the cells.

- 2) The value of sensory and spatial events is encoded by neurons in the VS (Pennartz et al., 2011; van der Meer & Redish, 2009). Experimental findings have clearly demonstrated that neurons in this area are involved in motivational and affective processing (Corbit & Balleine, 2003; Packard & McGaugh, 1992; Voorn, Vanderschuren, Groenewegen, Robbins, & Pennartz, 2004; Yin & Knowlton, 2006). Dopamine release in VS plays a key role in mediating affective control over motivated behaviour, and its dysregulation may contribute to disorders such as schizophrenia and drug addiction (Kurth-Nelson & Redish, 2012).
- 3) Hippocampal neurons do not form a metric map of the environment. Two experimental findings point to this conclusion. First, it has been observed that place cells can modify (remap) very rapidly their representation of space when objects or features of the environment are changed (Leutgeb et al., 2005; Rennó-Costa, Lisman, & Verschure, 2010). Should these cells define a spatial proximity relation, the remapping of one or more units would create a strong distortion of the cognitive map (since connections cannot be rewired instantly), for example with two distant places suddenly being connected. Second, during forward sweeps activity propagates first along one direction and then along the other (Johnson & Redish, 2007). If neurons were strongly connected to their neighbours, activity would propagate simultaneously in all directions like a wave-front.
- 4) The neuronal mechanisms for sensory processing and motor control are reused for planning and simulation (Kosslyn, Ganis, & Thompson, 2001). In the current implementation, the same visuomotor architectures support both overt action execution and imagery.
- 5) Spatial decision-making and mental simulations are the result of virtually ‘walking in the hippocampus’ (Hasselmo, 2009; Suddendorf, Addis, & Corballis, 2009) rather than being the result of isotropic activity diffusion. This concept is radically different from alternative models in the literature (Butz, Shirinov,

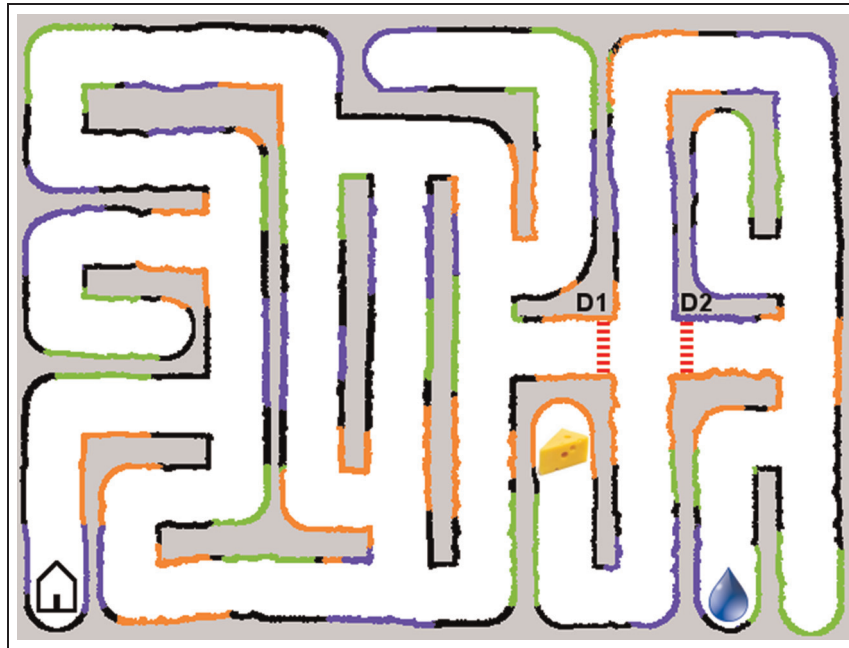


Figure 2. Representation of the maze with its most important features. In the bottom left corner is the Home position where the rat is replaced after each successful completion of the task. In the lower right part there are the two types of reward: cheese and water. D1 and D2 represent two doors that can be closed on demand and are used for example for detour tasks (see below). Coloured landmarks have been added to the walls in order to improve self-localization.

& Reif, 2010; Ivey, Bullock, & Grossberg, 2011; Lei, 1990; Martinet, Sheynikhovich, Benchenane, & Arleo, 2011; Toussaint, 2006) because it produces localized forward activity propagation (but see the Conclusions) and assigns a fundamental role to the motor and sensory areas in the planning process.

- 6) VS neurons can be excited and inhibited by pre-frontal (e.g., orbitofrontal) and limbic areas (Antzoulatos & Miller, 2011; Floresco, 2007). These modulations allow animals to switch between goals very rapidly, and to modify the assigned value of places and objects without rewiring their representations in the hippocampus.

3 Methods

3.1 Experimental set-up

In our set-up, a virtual rat can freely explore a complex maze (Figure 2) of the size of 3.20×4.80 m, which contains a home position (lower left corner), where the rat is repositioned at the beginning of every trial, and two locations with different types of rewards, respectively: food (cheese) and water. The explorable paths are delimited by walls that can be either black or of other different colours. These colours have been introduced in order to work as visual landmarks for place recognition and navigation.

The control system of the rat is implemented as a modular network of firing rate-based neurons (Figure 4).

The first module represents the visual areas and is composed of 121 neurons that encode the ‘coloured’ range map. More precisely, each of the 121 ‘lines of sight’ (Figure 3) that spawn the interval between -120° and $+120^\circ$ (referred to the head direction, every 2°) is encoded by four neurons, which represent the distance to the observed object and its colour in RGB format, respectively. *De facto*, this sensory vector may be interpreted as a union of a mono-dimensional depth map and a mono-dimensional image, both of width 121 and height 1 pixel.

The second module is the hippocampus and contains neurons that replicate place cells (green spheres in Figure 4). Each cell possesses a limited receptive field that covers only a small portion of the maze. As will be explained in more detail below, the extension of these fields is not determined by spatial but by visual features of the environment. For the sake of simplicity, this module has been modelled as a self-organizing Kohonen map (Kohonen, 2001). The number of effectively utilized units of this map depends on the characteristics of the environment and on the movement sequence, and increases during the exploration phase (see below).

As discussed earlier, place cells in this area are not connected to each other to form an internal map-like representation of the maze. Instead, they are connected to neurons in the VS, and receive projections from the sensory areas in an all-to-all fashion.

The input pattern p reaching the neurons in this area from the sensory module has the following form:

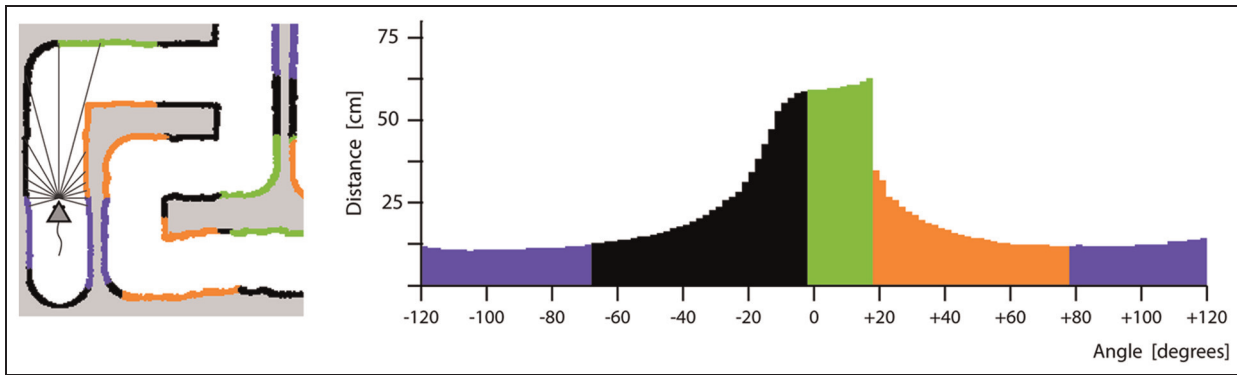


Figure 3. Left panel: detailed representation of the ‘lines of sight’. The rat (grey triangle) can see the environment with an angular aperture of 240° at a resolution of 2°. Each line of sight provides the depth and the colour of the intersected object. Right panel: the coloured range map obtained in the position indicated in the left panel. – 120° corresponds to the lower left-hand side of the rat, 0 to the forward direction and + 120 to the lower right-hand side.

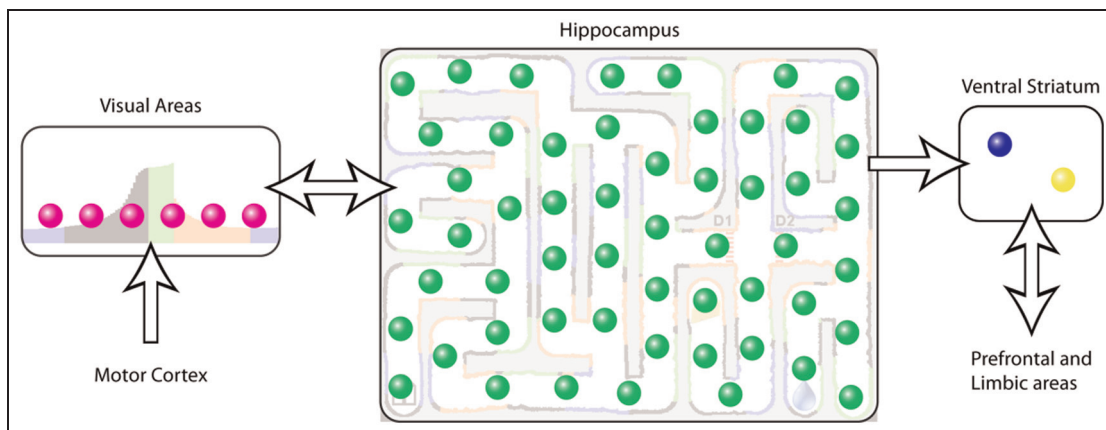


Figure 4. Representation of the complete architecture overlaid to the maze and to a range map. Each place cell in the Hippocampus (green spheres) is receptive for a small portion of the maze and projects to the value neurons in the ventral striatum. Visual neurons (purple spheres) encode the length of the line of sight and the colour of the intersected object. Neurons in the VS encode the value of each location through direct connections with hippocampal neurons.

$$p = (d_1, r_1, g_1, b_1, d_2, r_2, g_2, b_2, \dots, d_n, r_n, g_n, b_n) \quad (1)$$

where d_n is the normalized length of the n th line of sight ($n = 1, \dots, 121$), and r_n, g_n, b_n define the normalized colour of the object (i.e., the part of wall or the reward) seen in that specific direction.

The third module represents the VS and contains only a small number of neurons (in this particular implementation two, one for the food and one for the water), which encode the value of the current location of the maze.

In this case ‘value’ should be intended as the ‘return’ in TD-learning (Sutton, 1990; Sutton & Barto, 1998), i.e., an indication of the amount of reward an agent can expect to accumulate over the future, starting from that position, as opposed to ‘reward’, which is the amount of food or water the rat receives when reaching the foraging sites.

In this implementation, hippocampal and striatal neurons act in cooperation to provide spatial value information in the following way (Chersi & Pezzulo, 2012; Pezzulo, Rigoli, & Chersi, 2013). When the rat reaches a location in the maze, the corresponding place cell starts to fire at a high rate, signalling that the location has been recognized. This activation in turn elicits the firing of cells in the VS (due to the strong connectivity between the two areas) with an intensity that is proportional to the strength of their connections. In this view, the firing rate of a striatal neuron provides an indication of the proximity of the corresponding type of reward to the agent’s position (and thus the probability of obtaining it). In addition to this, the input to the VS from the prefrontal cortex and the limbic areas can produce a modulation of the activity of striatal neurons. For example, should the rat feel satiated with food, the striatal neuron encoding ‘cheese’ would receive a strong

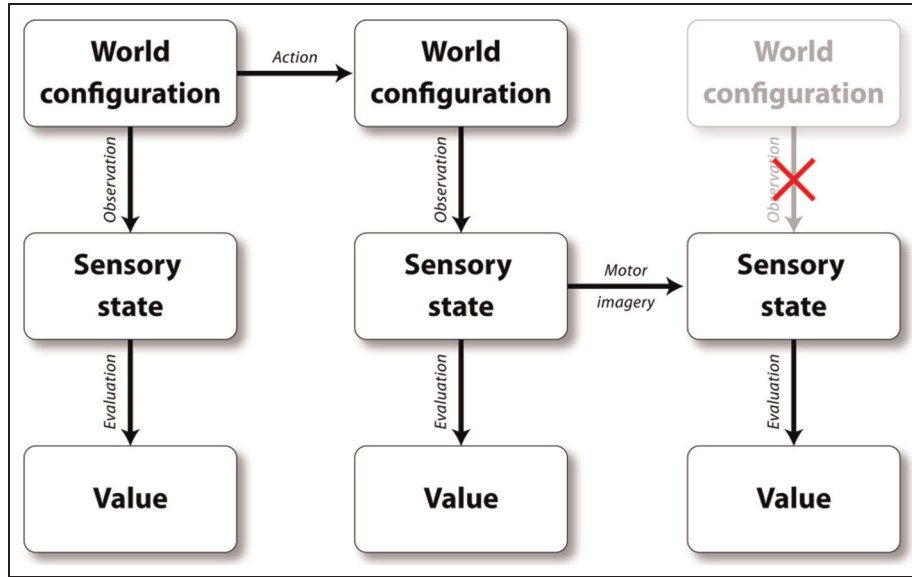


Figure 5. Schematic representation of the effects of actions (from left to middle column) on the configuration of the world (i.e., environment + agent) and the effect of motor imagery (from middle to right column) on the sensory state.

inhibitory signal from the limbic areas and thus be insensitive to the input deriving from the hippocampus (Corbit & Balleine, 2003). The projections from the hippocampus and limbic areas to the VS allow easy combination of information of different types in a single distributed representation.

The fourth module is the motor cortex. In the current implementation, we have not modelled the overt motor control aspect of this area in detail but instead we have focused on its motor imagery capabilities. In particular, we argue that the motor cortex is strongly involved in the transformation of sensory information by means of mental simulation (Figure 5). In this implementation, we exploited the (hypothesized) capacity of the motor cortex to transform a given visual input into another visual input exactly in the way overt motor actions effect real visual stimuli. For example, if a rat sees a wall at a certain distance in front of it, walking forward causes the wall to appear (on the retina) as getting closer to the animal. In a similar way, a clockwise rotation of the rat will produce a leftwards movement of all the objects on its retina. As we will show below, all these changes of the visual input can be obtained also by directly applying the corresponding motor-induced sensory transformation without the need to effectively execute the action (this is a hallmark of action simulation or emulation theories; Grush, 2004; Jeannerod, 2006).

In the present architecture, the above-mentioned transformations are not hardwired into the system but are learned through experience: as the rat explores the maze, it continuously receives visual inputs and is thus able to learn a mapping function between previous and subsequent sensory states as a function of the executed action. The motor imagery module has been modelled

as a multi-layer perceptron (MLP) network of sigmoidal units, where the input consists of the vector described by Equation 1 plus an additional part encoding the motor command in the form of step size Δd and rotation $\Delta\theta$:

$$p_{input}^+ = p_{input} \oplus (\Delta d, \Delta\theta) \quad (2)$$

The output of this network consists of a new vector of the form of Equation 1. In total, the MLP network contains $121 \times 4 + 2 = 486$ input neurons, 600 hidden and 484 output neurons.

The fifth module is the prefrontal cortex. In the current implementation, we sidestep most of the computations associated to this area, and only assign it the function of comparing the responses of the different striatal neurons in order to choose which action to execute. Furthermore, in the current implementation this area has not been implemented in form of a neural network, instead its functionalities are obtained algorithmically.

3.2 Mathematical details

In this implementation, each unit of the network represents a small subpopulation of neurons encoding information specific to the area it belongs: locations in the hippocampus, value in the VS and movements in the motor cortex.

The behaviour of each neuronal pool is described by a firing rate model with synaptic currents (Dayan & Abbott, 2001). This allows us to compactly represent complex interactions between excitatory and inhibitory neurons within pools and to take explicitly into account the dynamics of ionic currents and neurotransmitters.

The set of equations governing the behaviour of a single neuronal pool is the following:

$$\begin{cases} \tau_I \frac{dI_{syn}}{dt} = -I_{syn} + I_{ext} \\ \nu = g(I_{syn}) \\ \tau_J \frac{dJ}{dt} = -J + \nu \cdot (1 - J) \end{cases} \quad (3)$$

where I_{syn} is the total synaptic current, $\tau_I = 25$ ms, the corresponding time constant, ν is the firing rate, J is a current due to slow neurotransmitters that is used as an eligibility trace for learning (see below), $\tau_J = 500$ ms, the corresponding time constant, $g()$ is the current-to-firing rate (I - f) response function of a pool, which in this case has been modelled as:

$$\begin{cases} g(I) = g_{max} \cdot \tanh[\gamma(I - I_{thr})] & \text{for } I > I_{thr} \\ g(I) = 0 & \text{for } I < I_{thr} \end{cases} \quad (4)$$

where g_{max} determines the maximum firing rate, γ the steepness of the response function and I_{thr} is the firing threshold below which no response is present. In this implementation, $g_{max} = 120$ Hz, $\gamma = 3 \times 10^9$ A⁻¹ and $I_{thr} = 4 \times 10^{-10}$ A. All the parameters have been chosen in order to obtain a biologically realistic model.

3.3 Learning

In this architecture, there are three types of learning taking place within and between different areas of the network. In the following, we will describe these separately for the sake of clarity, but in fact they occur more or less simultaneously.

On one side, plastic connections between hippocampal and striatal neurons are updated utilizing a dopamine modulated eligibility trace-dependent rule (Sutton & Barto, 1998): when the animal reaches a foraging site

(i.e., water or cheese in the map), the corresponding VS neuron is activated and the network receives a reward in form of a dopamine increase, which enables the learning mechanism. The equations that govern the synaptic modification are the following:

$$\begin{aligned} \Delta w_{ik} &= \lambda J_i \nu_{VSk} DA \\ \tau_w \frac{dw_{ik}}{dt} &= -w_{ik} \end{aligned} \quad (5)$$

where w_{ik} is the weight between the i th neurons in the hippocampus and the k th neuron in the VS, λ is the learning rate, J_i is the slow current of the hippocampal neuron, ν_{VSk} is the firing rate of the VS neuron, DA is the dopamine increase, $\tau_w = 2000$ s is the weight decay time constant. The slow current J , which does not contribute to the neuronal firing rate, is used here as an eligibility trace for modifying synaptic weights between the place cells the animal has ‘visited’ earlier and striatal neurons.

As training proceeds, place cells become more strongly connected to the VS neurons in a way that is directly proportional to their proximity to the goal place they encode (because the eligibility trace is more intense closer to the reward site). The final result of this learning procedure is the formation of neural maps with a weight gradient leading towards the foraging locations (Figure 6).

A different type of learning takes place in the sensory-motor circuit. As stated above, the main role of this circuit in this implementation is to produce mental simulations of actions related to navigation, i.e., of forward movements and rotations. Since these simulations in practice consist in creating the visual pattern that ‘follows’ a given visual input, this problem can be rephrased in terms of learning the laws of optic flow utilizing couples of visual patterns and subsequently applying these rules to the input patterns:

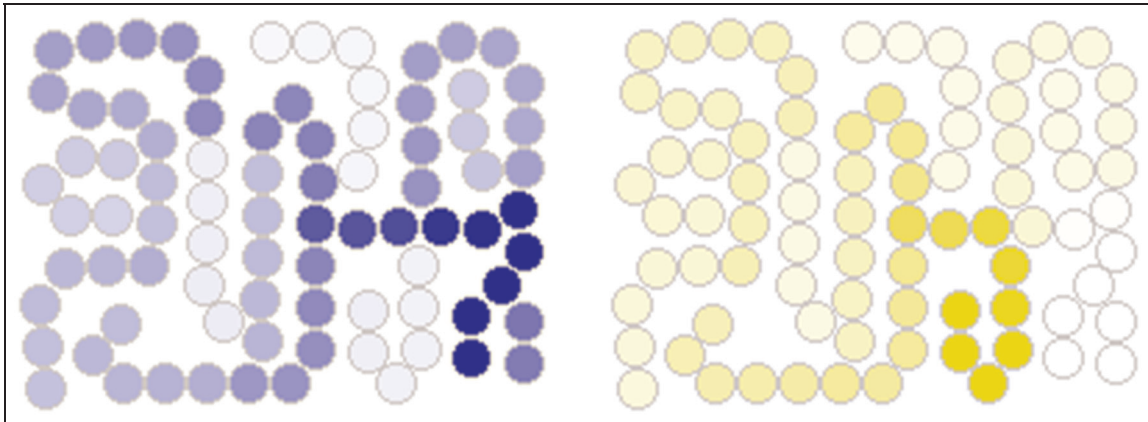


Figure 6. Connections’ strength between the place cells representing the maze in Figure 2 and the two striatal neurons encoding the different rewards (yellow=cheese, blue=water) after 30 trials. Darker colours mean stronger weights. Due to the employed learning rule, these weights form a gradient that increases in direction of the reward locations.

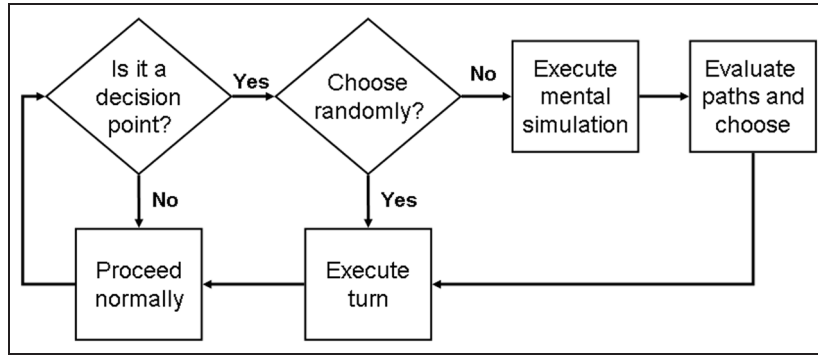


Figure 7. Schematic representation of the decision-making process.

$$\Phi = \mathbf{P}_{\text{output}} / \mathbf{P}_{\text{input}}^+$$

$$\mathbf{p}_{\text{output}} = \Phi(\mathbf{p}_{\text{input}}^+) \quad (6)$$

$$\mathbf{p}_{\text{output}} = \Phi(\mathbf{p}_{\text{input}}, \Delta d, \Delta \theta)$$

where Φ is the optic flow, i.e., the sought transformation function, which we have indicated as the result of an abstract quotient between visual patterns of the form of Equations 1 and 2. Note that the step size Δd and the rotation $\Delta \theta$ are not visual features and are treated as parameters of the transformation.

Instead of providing this capability as a hardwired property of the system, we chose to let the agent learn it through experience. To this aim, we initially isolated part of the maze by closing the first door (D1 in Figure 2) and let the rat wander freely collecting data patterns to train the neural network.

While learning the optic flow transformations, the rat also utilizes the sensory input to train the place cells that are thereafter used for self-localization.

Visual input patterns are sampled approximately at regular space intervals and are stored in the hippocampus layer (a SOM map) according to the Kohonen learning rule. Initially, the weight vectors of the neurons are set to small random values. When a training pattern $p(t)$ (Equation 1) is fed to the network, its Euclidean distance to all weight vectors is computed. The neuron whose weight vector is most similar to the input is called the best matching unit (BMU). The weights of the BMU and neurons close to it in the SOM lattice are adjusted towards the input vector. The magnitude of the change decreases with time and with distance (within the lattice) from the BMU. The update formula for a neuron with weight vector $W(t)$ is

$$W(t+1) = W(t) + \Theta(i, j, t) \alpha(t) [p(t) - W(t)] \quad (7)$$

where t is the time step, $p(t)$ is the current training pattern, i is the index of the BMU for $p(t)$, $\alpha(t)$ is a monotonically decreasing learning coefficient, j an index that spans over all neurons for every value of t . The neighbourhood function $\Theta(i, j, t)$ depends on the lattice distance between neuron i and neuron j . It is equal to 1

when $i=j$ and its value decreases when the distance between the neuron i and neuron j on the lattice increases.

In order to obtain a higher disambiguation capacity of the network, we chose to add an extra term that penalizes the matching of a previously BMU when the animal moves away from the previous receptive field for more than a threshold value (in our case 2.5 cm).

3.4 Mental simulation and decision-making

The ultimate goal of the rat is to obtain as much reward as possible. To this aim, while running around in the maze, at each intersection it has to choose the direction that leads most rapidly to the foraging site providing the desired reward.

In our model, this decision is the result of two contributions (Figure 7): a random choice generator and a goal-directed decision mechanism. The first one has been introduced to generate variability in the decisions and thus an exploratory behaviour of the animal.

The second one uses a model-based approach to determine the best choice to make at a given decision point. It utilizes motor imagery to construct and evaluate all possible action sequences compatible with the current position (e.g., moving either along the right or along the left arm at a crossing).

More specifically, when the rat reaches a crossing point, it utilizes the available visual input as a starting point for the generation of subsequent visual patterns by applying repeatedly the parametric optic flow transformation with the parameters being the direction and step size of the intended movement (Figure 8). At every simulation step the new synthetic sensory state is passed to the hippocampus, which, treating it as an ordinary input, classifies it (i.e., a specific place cell responds) and in turn causes the activation of the striatal neurons according to previously learned associations (Figure 8). The created pattern is then taken as new starting point for the generation of the subsequent pattern.

It is important to note that neither the hippocampus nor the striatum are ‘aware’ that the sensory input is not real but instead mentally constructed as, for what

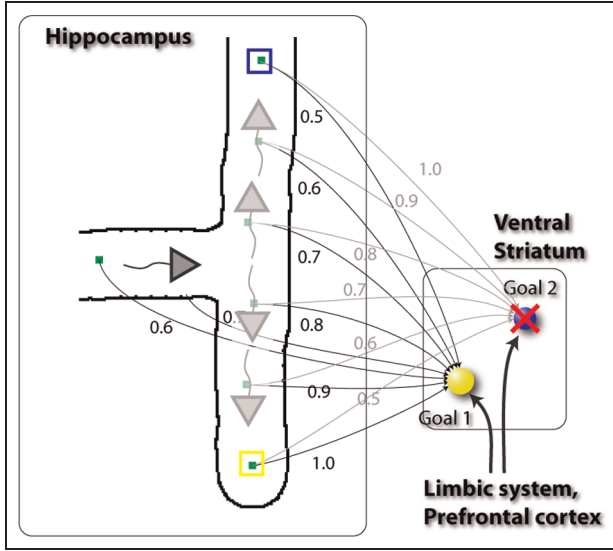


Figure 8. Schematic representation of the proposed decision mechanism based on mental simulation: at a T-crossing the rat imagines itself first moving along the right arm then along the left arm. Activity from hippocampal neurons propagates to the Striatum eliciting a response that is proportional to the strength of the connections, thus providing a measurement of the value of each choice.

concerns the format, the new input is undistinguishable from the real signals.

We want to underline the fact that the process of creation of synthetic sensory inputs cannot be repeated indefinitely because at every iteration small errors are introduced, which add up, eventually producing artifacts that impair the correct recognition (Figure 9). The solution that we have introduced consists in repeating the mental imagery until the receptive field of a subsequent place cell is reached. At this point, the visual pattern associated to the new place cell is taken as the reference pattern for creating new visual sequences. In this implementation, we have chosen to produce forward sweeps as far as three place cells away from the current position, a length that appeared to be sufficient to produce a good estimation of the value of a path (but see the Discussion).

The total value U of a path is calculated as the average of all the activations ν_i of the striatal neuron representing the desired goal (food or water) along the path.

$$U = \frac{1}{N} \cdot \sum_{i=1}^N \nu_i \quad (8)$$

where ν_i is the response of the striatal neuron and N is the number of simulated steps (where one step is equivalent to the size of the receptive field).

The decision is taken following the direction that produced the highest striatal response. The complete algorithm thus becomes:

if $rand < \theta_{rand}$	→ randomly choose a direction of movement
else	→ calculate U_i for each direction i and find U_i^{max}
if $U_i^{max} < \theta_U$	→ randomly choose a direction of movement
else	→ move in the direction of U_i^{max}

(9)

where $rand$ is a randomly generated number, θ_{rand} is the threshold for random movement generation, U_i is the value of the i th path calculated according to Equation 8, θ_U is the second threshold for random movement generation.

Decisions about the direction to take are initially completely random as all U_i are very small, but as learning proceeds the neural responses become stronger and this begins to influence the choices favouring the direction of highest U value.

4 Results

In the following, we first report results concerning the outcome of mental simulation and then concerning the performance of the agent.

4.1 Mental simulation

In the current experiment the MLP was trained with 6000 couples of patterns collected in the explorable part of the maze (approximately two thirds of the whole) at distances between $\Delta d = 2.5$ and $\Delta d = 7.5$ cm and rotation angles between $\Delta \theta = -10^\circ$ and $\Delta \theta = +10^\circ$.

The left panel of Figure 9 shows the reconstruction accuracy of the network for 5, 10, 15, 20 and 25 cm of simulated movement, before training, after 100, 200 and 1000 epochs. A 100% accuracy means that network outputs are identical to the desired responses, while a 0% accuracy corresponds to patterns that are maximally different (note that the error has a maximum bound because all neuronal outputs are comprised between 0 and 1).

To verify the generalization capabilities of the proposed model, after the initial training phase, the rat was positioned in the unexplored part of the maze and we tested the network's ability to reconstruct untrained patterns. Results are shown in the right panel of Figure 9.

Note that as soon as the rat enters the unknown part of the maze, it starts to train the visuo-motor circuit with the newly collected data thus rapidly increasing its reconstruction performance.

4.2 Goal-directed behaviour

After learning the visual transformations, the model was validated in the multiple T-maze shown in

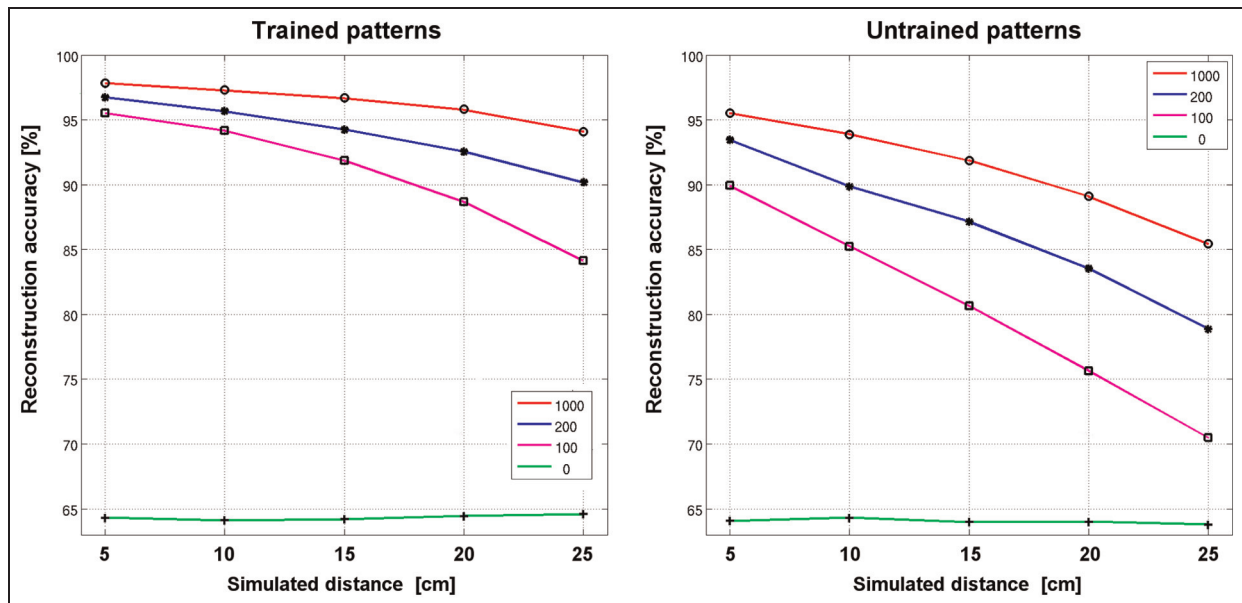


Figure 9. Left panel: performance of the multi-layer perceptron in simulating 5, 10, 15, 20 and 25 cm forward, before training (green curve), after 100 (purple), 200 (blue) and 1000 training epochs (red curve). The right panel shows measurements similar to the left panel but using as test data visual patterns collected in the unexplored part of the maze. A 0 reconstruction accuracy corresponds to patterns being maximally different.

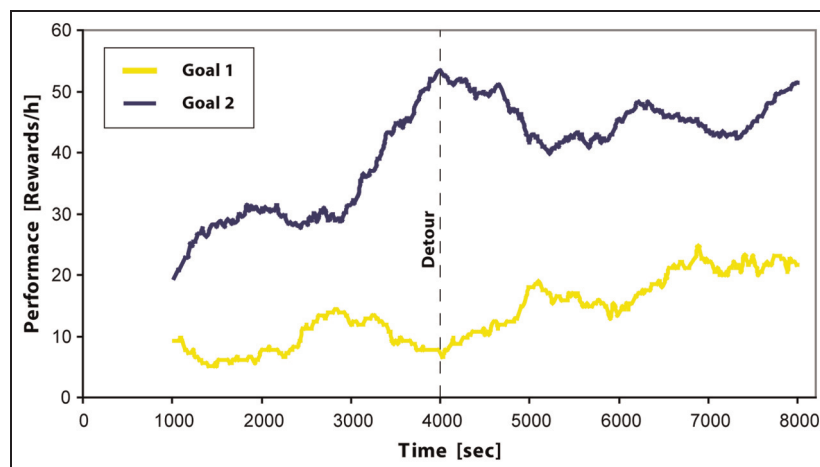


Figure 10. Average performance of the rat measured as the number of rewards received per hour (calculated as the moving average over 1000 s). The yellow and blue curves represent the rat's performance in reaching goal 1 (the cheese) and 2 (the water), respectively. The dashed line represents the moment in which the second door is closed, thus forcing the rat to take a detour in order to reach goal 2.

Figure 2. Each trial started with the rat being placed in the home position (bottom left corner) and then allowed to move freely around the maze. Each time it found either cheese or water, it received a reward in form of a dopamine input to the network and the trial ended.

After 4000 s, we closed the second door of the maze (D2 in Figure 2), thus forcing the rat to choose a detour to reach the second goal (i.e., the water).

Figure 10 shows the performance of the rat in reaching the two foraging sites. Here the goal was to reach the

water location, thus the mental simulation, when possible, was utilized solely for this aim. Moreover, the probability of random decision at each crossing was set to 20%.

It is worth to explain how motor imagery works in the detour situation. When the rat starts from the left part of the maze and reaches the decision point in front of the closed door, it sees a 'standard' T-crossing situation with one arm on the left and one arm on the right. It will thus simulate moving along the only two possible directions, generating two sequences of new visual

patterns starting from its current state. Note that a mechanism that strongly relies on sensory information instead of on learned graph-like mental representations, can easily cope with dynamic and uncertain environments (Pezzulo et al., 2013).

As can be seen the performance of the rat increases in time for both goals. This is because the first part of the path is in common for the two. Moreover, the introduction of a block (door 2 is closed at $t=4000$ s) temporarily degrades the performance concerning goal 2. This is because, although the rat continues to be able to perform mental simulations also along the longer path, the value encoded by the corresponding VS neurons is below threshold thus eliciting in the beginning random responses (Equation 9).

5 Conclusions and discussion

We proposed a biologically inspired model of the hippocampus-VS circuit involved in goal-directed decision-making, which belongs to the broader category of ‘model-based’ reinforcement learning models. It includes a mechanism for forward search and a mechanism of evaluation of the covert expectations of reward produced by the search (Daw, Niv, & Dayan, 2005; Pezzulo et al., 2013; Pezzulo & Rigoli, 2011; Solway & Botvinick, 2012). The model-based approach relies on a ‘forward model’ of the environment (akin to the idea of a cognitive map; Tolman, 1948) to mentally simulate possible navigation paths and their outcomes. A core assumption of our model is that at decision points animals simulate the outcome of possible actions by virtually ‘walking in the hippocampus’, meaning that actions and decisions are now taken by ‘detaching’ the brain from the real world and utilizing exclusively the acquired knowledge about the environment and about the physical laws that govern it. This, however, is not obtained by using a different mechanism or a specific part of the brain that is dedicated to simulations, but instead it involves exactly the same areas that are active during overt action execution. As a result simulations seem as real and embodied to these brain areas as the physical actions at the point that it is possible to observe neural responses that are undistinguishable from normal activity (e.g., the ‘forward sweeps’). This characteristic distinguishes our model from related ones that implement predictions and planning in the navigation domain using diffusion methods or using cells that encode abstract action-outcome representations (Butz et al., 2010; Ivey et al., 2011; Lei, 1990; Martinet et al., 2011; Toussaint, 2006).

5.1 Comparison with previous work

After the seminal article by Daw et al. (2005) a recent stream of research, combining empirical and computational methods, is exploring the mechanisms

of goal-directed choice in terms of model-based reinforcement learning. One of the early and popular examples of model-based systems was the DYNA architecture (Sutton, 1990), which has been also recently re-proposed as a cognitive mechanism for explaining the shift between habitual and goal-directed systems and retrospective revaluation (Gershman, Markman, & Otto, 2012). Different from the proposed model, the DYNA architecture uses mental simulation only to train (off-line) the habitual system but not during choice. Thus the DYNA architecture would not be able to model the so-called ‘forward sweeps’ in the hippocampus at decision points. We do not see our model and the DYNA system as contradictory; rather, they could model two complementary roles of model-based computations: supporting on-line goal-directed choice (as in our model) and off-line training of the habitual system (as in the DYNA system). These could correspond to hippocampal forward sweeps and ‘replay’ at decision points and at rest (or when the rat is asleep), respectively. A more recent example of model-based approach to goal-directed choice was advanced by Solway and Botvinick (2012). This model is able to reproduce numerous aspects of goal-directed computations such as the sensitivity to outcome devaluation and the possibility to readapt to changed environmental conditions (required, e.g., for detour) by using a combination of backward and forward search. Instead, in keeping with evidence of forward sweeps in the hippocampus, our model uses only forward search. It remains to be elucidated under what conditions the brain uses forward and/or backward search computations (that can be both linked to similar model-based computations).

In comparison to the model of Erdem and Hasselmo (2012), our architecture has the advantage that it can naturally simulate paths of any shape (not just straight lines) and circumvent distant obstacles because the generation of trajectories is guided step by step by motor responses to sensory stimuli (even though they are self-generated). Moreover, it does not need a graph-like representation of the environment in the prefrontal cortex for reward diffusion as it utilizes the value encoding capability of the VS.

To conclude, in spite of some evident limitation such as the restricted capability of mental simulation due to the accumulation of errors, the lack of realistic model-free behaviour, the simplified value-based computation mechanism and the approximated mapping to anatomical structures, the presented architecture could constitute a building block for more sophisticated functions linked to hippocampal processing, such as imagination, mental time travel and constructive memory (Buckner & Carroll, 2007; Hassabis et al., 2007). The neural bases of such abilities are nowadays deeply studied, and accumulating evidence indicates that episodic memories (traditionally linked to hippocampal function) can be

flexibly reused and reassembled in novel ways (Schacter et al., 2012). It is thus plausible that constructive memory reuses mechanisms similar to forward sweeps in a more flexible way to reassemble episodic memories so as to generate (and possibly evaluate) future and never-experienced scenarios. Living organisms that can use this mechanism (along with related ones such as visual and motor imagery) to prepare dealing with future and novel events have significant adaptive advantages compared with other animals that can only respond to current stimuli.

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References

- Antzoulatos, E. G., & Miller, E. K. (2011). Differences between neural activity in prefrontal cortex and striatum during learning of novel abstract categories. *Neuron*, 71(2), 243–249.
- Bar, M. (2009). The proactive brain: Memory for predictions. *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences*, 364(1521), 1235–1243.
- Buckner, R. L., & Carroll, D. C. (2007). Self-projection and the brain. *Trends in Cognitive Sciences*, 11(2), 49–57.
- Butz, M. V., Shirinov, E., & Reif, K. L. (2010). Self-organizing sensorimotor maps plus internal motivations yield animal-like behavior. *Adaptive Behavior*, 18(3–4), 315–337.
- Chersi, F., & Pezzulo, G. (2012). Using hippocampal-striatal loops for spatial navigation and goal-directed decision-making. *Cognitive Processing*, 13 Suppl 1, S125–129.
- Corbit, L. H., & Balleine, B. W. (2003). The role of prelimbic cortex in instrumental conditioning. *Behavioural Brain Research*, 146, 145–157.
- Daw, N. D., Niv, Y., & Dayan, P. (2005). Uncertainty-based competition between prefrontal and dorsolateral striatal systems for behavioral control. *Nature Neuroscience*, 8(12), 1704–1711.
- Dayan, P., & Abbott, L. F. (2001). *Theoretical Neuroscience. Computational and Mathematical Modelling of Neural Systems*. Cambridge, MA: MIT Press.
- Dragoi, G., & Tonegawa, S. (2011). Preplay of future place cell sequences by hippocampal cellular assemblies. *Nature*, 469(7330), 397–401.
- Erdem, U. M., & Hasselmo, M. (2012). A goal-directed spatial navigation model using forward trajectory planning based on grid cells. *The European Journal of Neuroscience*, 35(6), 916–931.
- Floresco, S. B. (2007). Dopaminergic regulation of limbic-striatal interplay. *Journal of Psychiatry & Neuroscience*, 32(6), 400–411.
- Gershman, S. J., Markman, A. B., & Otto, A. R. (2012). Retrospective revaluation in sequential decision making: A tale of two systems. *Journal of Experimental Psychology. General*. 2012 Dec 10. [Epub ahead of print].
- Grush, R. (2004). The emulation theory of representation: Motor control, imagery, and perception. *The Behavioral and Brain Sciences*, 27(3), 377–396; discussion 396–442.
- Gupta, A. S., van der Meer, M. A. A., Touretzky, D. S., & Redish, A. D. (2010). Hippocampal replay is not a simple function of experience. *Neuron*, 65(5), 695–705.
- Hassabis, D., Kumaran, D., & Maguire, E. A. (2007). Using imagination to understand the neural basis of episodic memory. *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience*, 27(52), 14365–14374.
- Hasselmo, M. E. (2009). A model of episodic memory: Mental time travel along encoded trajectories using grid cells. *Neurobiology of Learning and Memory*, 92(4), 559–573.
- Ivey, R., Bullock, D., & Grossberg, S. (2011). A neuro-morphic model of spatial lookahead planning. *Neural Networks: the Official Journal of the International Neural Network Society*, 24(3), 257–266.
- Jeannerod, M. (2006). *Motor cognition: What actions tell the self*. Oxford, UK: Oxford University Press.
- Johnson, A., & Redish, A. D. (2007). Neural ensembles in CA3 transiently encode paths forward of the animal at a decision point. *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience*, 27(45), 12176–12189.
- Kohonen, T. (2001). *Self-organizing maps*. Heidelberg, Germany: Springer.
- Kosslyn, S. M., Ganis, G., & Thompson, W. L. (2001). Neural foundations of imagery. *Nature Reviews Neuroscience*, 2(9), 635–642.
- Kurth-Nelson, Z., & Redish, A. D. (2012). Modeling decision-making systems in addiction. In B. Gutkin & S. H. Ahmed (Eds.), *Computational Neuroscience of Drug Addiction* (pp. 163–187). New York: Springer. Retrieved from http://link.springer.com/chapter/10.1007/978-1-4614-0751-5_6.
- Lei, G. (1990). A neuron model with fluid properties for solving labyrinthian puzzle. *Biological Cybernetics*, 64(1), 61–67.
- Leutgeb, S., Leutgeb, J. K., Barnes, C. A., Moser, E. I., McNaughton, B. L., & Moser, M.-B. (2005). Independent codes for spatial and episodic memory in hippocampal neuronal ensembles. *Science*, 309(5734), 619–623.
- Lisman, J., & Redish, A. D. (2009). Prediction, sequences and the hippocampus. *Philosophical transactions of the Royal Society of London. Series B, Biological Sciences*, 364(1521), 1193–1201.
- Martinet, L.-E., Sheynikhovich, D., Benchenane, K., & Arleo, A. (2011). Spatial learning and action planning in a prefrontal cortical network model. *PLoS Computational Biology*, 7(5), e1002045.
- Moulton, S. T., & Kosslyn, S. M. (2009). Imagining predictions: mental imagery as mental emulation. *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences*, 364(1521), 1273–1280.
- Muller, R. U., Kubie, J. L., Bostock, E. M., Taube, J. S., & Quirk, G. J. (1991). Spatial firing correlates of neurons in the hippocampal formation of freely moving rats. In J. Paillard (Ed.), *Brain and Space*. New York: Oxford University Press.

- O'Keefe, J., & Dostrovsky, J. (1971). The hippocampus as a spatial map. Preliminary evidence from unit activity in the freely-moving rat. *Brain Research*, 34(1), 171–175.
- O'Keefe, J., & Burgess, N. (2005). Dual phase and rate coding in hippocampal place cells: theoretical significance and relationship to entorhinal grid cells. *Hippocampus*, 15(7), 853–866.
- Packard, M. G., & McGaugh, J. L. (1992). Double dissociation of fornix and caudate nucleus lesions on acquisition of two water maze tasks: Further evidence for multiple memory systems. *Behavioral Neuroscience*, 106(3), 439–446.
- Papale, A. E., Stott, J. J., Powell, N. J., Regier, P. S., & Redish, A. D. (2012). Interactions between deliberation and delay-discounting in rats. *Cognitive, Affective & Behavioral Neuroscience*, 12(3), 513–526.
- Pastalkova, E., Itskov, V., Amarasingham, A., & Buzsáki, G. (2008). Internally generated cell assembly sequences in the rat hippocampus. *Science (New York, N.Y.)*, 321(5894), 1322–1327.
- Pennartz, C. M. A., Ito, R., Verschure, P. F. M. J., Battaglia, F. P., & Robbins, T. W. (2011). The hippocampal-striatal axis in learning, prediction and goal-directed behavior. *Trends in Neurosciences*, 34(10), 548–559.
- Pezzulo, G., Butz, M. V., Castelfranchi, C., & Falcone, R. (2008). *The challenge of anticipation – A unifying framework for the analysis and design of artificial*. Berlin, Germany: Springer.
- Pezzulo, G., & Castelfranchi, C. (2007). The symbol detachment problem. *Cognitive Processing*, 8(2), 115–131.
- Pezzulo, G., & Rigoli, F. (2011). The value of foresight: How prospection affects decision-making. *Frontiers in Neuroscience*, 5, 79.
- Pezzulo, G., Rigoli, F., & Chersi, F. (2013). The mixed instrumental controller: Using value of information to combine habitual choice and mental simulation. *Frontiers in Psychology*, 4, 92.
- Redish, A. D. (2012). Search processes and hippocampus. In P. M. Todd, T. T. Hills, & T. W. Robbins (Eds.), *Cognitive search: Evolution, algorithms, and the brain* (pp. 81–95). Strüngmann Forum Reports. Cambridge, MA: MIT Press.
- Rennó-Costa, C., Lisman, J. E., & Verschure, P. F. M. J. (2010). The mechanism of rate remapping in the dentate gyrus. *Neuron*, 68(6), 1051–1058.
- Schacter, D. L., Addis, D. R., Hassabis, D., Martin, V. C., Spreng, R. N., & Szpunar, K. K. (2012). The future of memory: remembering, imagining, and the brain. *Neuron*, 76(4), 677–694.
- Solway, A., & Botvinick, M. M. (2012). Goal-directed decision making as probabilistic inference: a computational framework and potential neural correlates. *Psychological Review*, 119(1), 120–154.
- Suddendorf, T., Addis, D. R., & Corballis, M. C. (2009). Mental time travel and the shaping of the human mind. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 364(1521), 1317–1324.
- Sutton, R. S. (1990). Integrated architectures for learning, planning, and reacting based on approximating dynamic programming. In *Proceedings of the Seventh International Conference on Machine Learning* (pp. 216–224). Burlington, MA: Morgan Kaufmann.
- Sutton, R. S., & Barto, A. G. (1998). *Reinforcement learning: An introduction*. Cambridge, MA: MIT Press.
- Tolman, E. C. (1948). Cognitive maps in animals and man. *Psychological Review*, 55, 189–208.
- Toussaint, M. (2006). A sensorimotor map: Modulating lateral interactions for anticipation and planning. *Neural Computation*, 18(5), 1132–1155.
- Van der Meer, M. A. A., & Redish, A. D. (2009). Covert expectation-of-reward in rat ventral striatum at decision points. *Frontiers in Integrative Neuroscience*, 3, 1.
- Van der Meer, M. A. A., & Redish, A. D. (2010). Expectancies in decision making, reinforcement learning, and ventral striatum. *Frontiers in Neuroscience*, 4, 6.
- Voorn, P., Vanderschuren, L. J., Groenewegen, H. J., Robbins, T. W., & Pennartz, C. M. (2004). Putting a spin on the dorsal-ventral divide of the striatum. *Trends in Neuroscience*, 27(8), 468–474.
- Yin, H. H., & Knowlton, B. J. (2006). The role of the basal ganglia in habit formation. *Nat Rev Neurosci*, 7(6), 464–476.

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